- Periodic and Non-periodic signals

* Non-periodic signals are also called Aperiodic signals.

- A signal is said to be periodic if it repeates itself after a regular interval of time.

$$\boxed{x(t) = x(t \pm nT)}$$, $T \rightarrow$ Time period. (fundamental period)

$n \rightarrow$ integer.

Example : $T=5$, $x(t) = x(t+5)$

$x(t+5) = x(t+10)$

- Fundamental frequency :

$$f_0 = \frac{1}{T} \quad Hz$$

* For DTS

$$x[n] = x[n \pm \underbrace{m}_{\text{integer}} \underbrace{N}_{\text{fundamental period. (an integer)}}]$$

- Calculation of fundamental period (T)

↳ $x(t) = x(t+nT)$

let $n=1$

↳ $x(t) = x(t+T)$

Example : $x(t) = A_0 e^{j\omega t}$

↳ Step. 1 $\quad x(t+T) = A_0 e^{j\omega(t+T)}$

↳ Step. 2 $\quad \cancel{A_0} e^{j\omega t} = \cancel{A_0} e^{j\omega(t+T)}$

$\cancel{e^{j\omega t}} = \cancel{e^{j\omega t}} . e^{j\omega T}$

↳ Step. 3 $\quad e^{j\omega T} = 1$

↳ Using Euler's formula :

$$\boxed{\begin{array}{l} e^{jx} = \cos x + j\sin x \\ \text{let } x = 2\pi k \\ e^{j2\pi k} = \cos 2\pi k + j\sin 2\pi k \end{array}}$$

$$e^{j2\pi k} = \underbrace{\cos 2\pi k}_{1} + j\underbrace{\sin 2\pi k}_{0}$$

$$e^{j2\pi k} = 1$$

$$\therefore \quad e^{j\omega T} = 1 = e^{j2\pi k}$$

$$\omega T = 2\pi k$$

$$\boxed{T = \frac{2\pi}{\omega} . k}$$

if $k=1$

$$\boxed{T = \frac{2\pi}{\omega}}$$

Note: $x(t) = x_1(t) \longrightarrow$ Non-composite signals.

$x(t) = x_1(t) + x_2(t) \longrightarrow$ These are Composite signals

* Calculation of fundamental period (Composite Signals)

$\hookrightarrow x(t) = x_1(t) + x_2(t)$

$\hookrightarrow x(t) = x_1(t) * x_2(t)$

imp.

* Sum (or product) of two or more periodic signals may or may not be periodic.

STEP. 1 : Determine fundamental period of individual signals.

$$x(t) = \underbrace{x_1(t)}_{\text{fundamental period be } T_1} + \underbrace{x_2(t)}_{T_2}$$

STEP. 2 : Find the ratio of fundamental period of 1st signal to fundamental period of every other signal.

$\hookrightarrow$ Here, ratio $\frac{T_1}{T_2}$. If we had $x_3(t)$ then we need to find $\frac{T_1}{T_3}$ too.

$$\left(\frac{T_1}{T_2}, \frac{T_1}{T_3}\right)$$

STEP.3 : If the ratio's obtained in previous step are Rational, then the composite signal is periodic.

STEP.4 : This step is performed only if Ratio's are ~~peri~~ Rational. (all ratio's)

$$\boxed{T = LCM\,(T_1, T_2, \text{----})}$$

(fundamental period)

Example : $x(t) = \sin 4\pi t + \cos 2t$

$\hookrightarrow$ $x_1(t) = \sin 4\pi t$ $\quad$ $x_2(t) = \cos 2t$

Step.1 $\quad T_1 = \dfrac{2\pi}{\omega_1} = \dfrac{2\pi}{4\pi} = \dfrac{1}{2}$ $\qquad$ $T_2 = \dfrac{2\pi}{\omega_2} = \dfrac{2\pi}{2} = \pi$

Step.2 $\quad \dfrac{T_1}{T_2} = \dfrac{1/2}{\pi} = \dfrac{1}{2\pi}$

Step.3 $\quad$ The Ratio is irrational. Hence the composite signal is Non-periodic.

Example : $x(t) = \underbrace{\boxed{5\cos \pi t}}_{x_1(t)} + \underbrace{\boxed{\sin 3\pi t}}_{x_2(t)}$

Step.1 $\quad T_1 = \dfrac{2\pi}{\pi} = 2$, $T_2 = \dfrac{2\pi}{3\pi} = \dfrac{2}{3}$ $\quad$ (Fundamental periods of individual signals)

Step.2 $\quad \dfrac{T_1}{T_2} = \dfrac{6}{2} = 3$ (Rational)

Step.3 $\quad$ Fundamental period of $x(t) = T$

$$T = LCM\left(2, \frac{2}{3}\right)$$

* LCM of fractions

V. Imp.

$$\boxed{\text{LCM of fractions} = \frac{\text{LCM of Numerators}}{\text{HCF of denominators}}}$$

$$T = \text{LCM}\left(\frac{2}{1}, \frac{2}{3}\right)$$

$$\rightarrow T = \frac{\text{LCM}(2,2)}{\text{HCF}(1,3)} = \frac{2}{1}$$

$$\boxed{\therefore \text{ fundamental period}, T = 2}$$

Ques: Calculate fondamental period and fundamental frequency of following signals.

(i) $x(t) = \sin^2(4\pi t)$

$$\cos 2\theta = 1 - 2\sin^2\theta$$

or $\sin^2\theta = \dfrac{1-\cos 2\theta}{2}$; $\theta = 4\pi t$

$$\therefore \sin^2 4\pi t = \frac{1}{2}\left[1 - \cos 8\pi t\right]$$

$$\therefore x(t) = \frac{1}{2} - \frac{\cos 8\pi t}{2}$$

DC value
→ periodic
→ T → undefined

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{8\pi}$$

$$\boxed{T = \tfrac{1}{4} \text{ seconds} = 0.25 \text{ sec}}$$

and this our fondamental period

⇒ fondamental frequency, $f = \frac{1}{T} = 4\ Hz$

ii) $x(t) = \underbrace{\sin 6\pi t}_{x_1(t)} + \underbrace{\cos 5\pi t}_{x_2(t)}$

Step-1 $T_1 = \frac{2\pi}{6\pi} = \frac{1}{3}$, $T_2 = \frac{2\pi}{5\pi} = \frac{2}{5}$

Step-2 $\frac{T_1}{T_2} = \frac{2/6}{2/5} = \frac{5}{6} = 0.8333\overline{33}$ (Repeating)

Hence Rational.

Step-3 $T = LCM(T_1, T_2) = LCM\left(\frac{1}{3}, \frac{2}{5}\right)$

$$\boxed{T = \frac{LCM(1,2)}{HCF(3,5)} = \frac{2}{1}}$$

(fundamental period) = 2 sec.

$\Rightarrow f = 1/T = \frac{1}{2} = 0.5\ Hz$